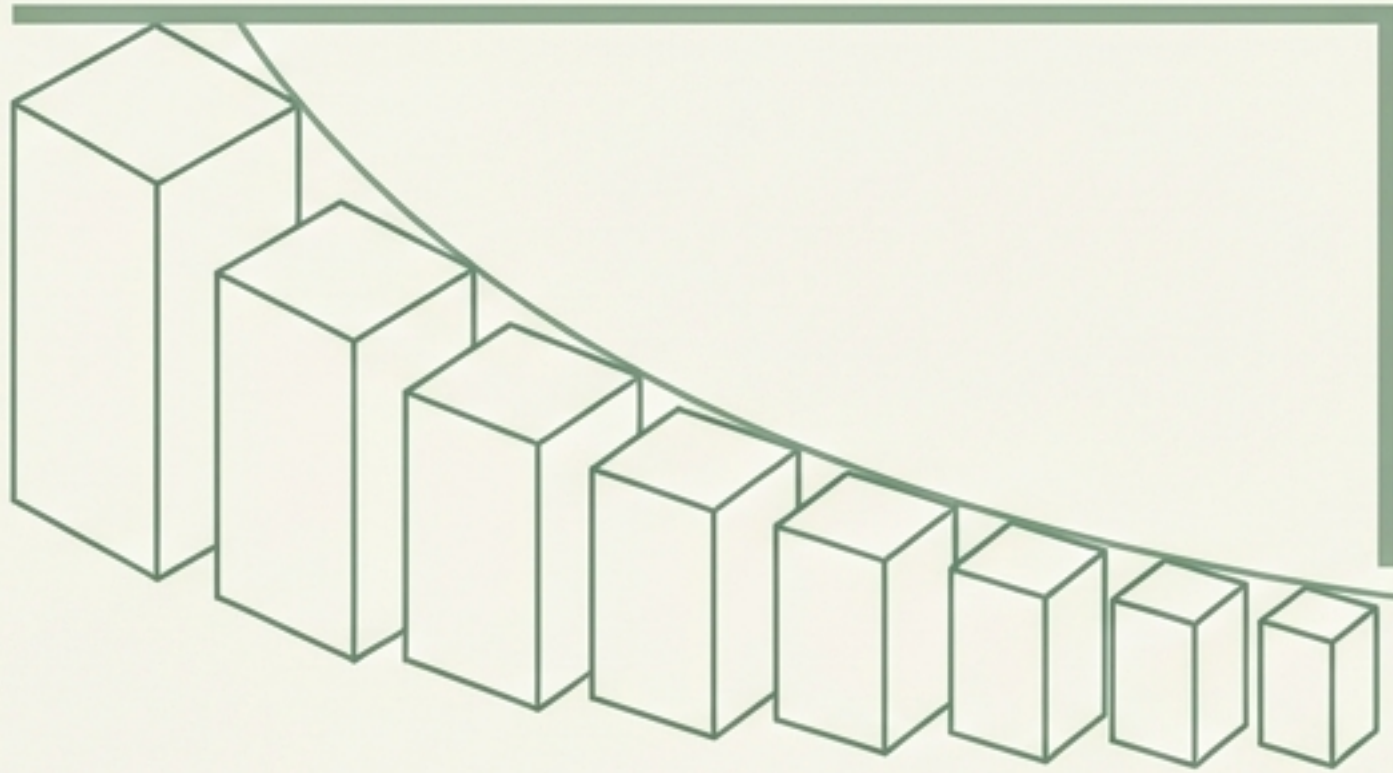


Algorithmic Calculus: Decoding Infinite Series

A diagnostic playbook for convergence,
mathematical tests, and foundational expansions.

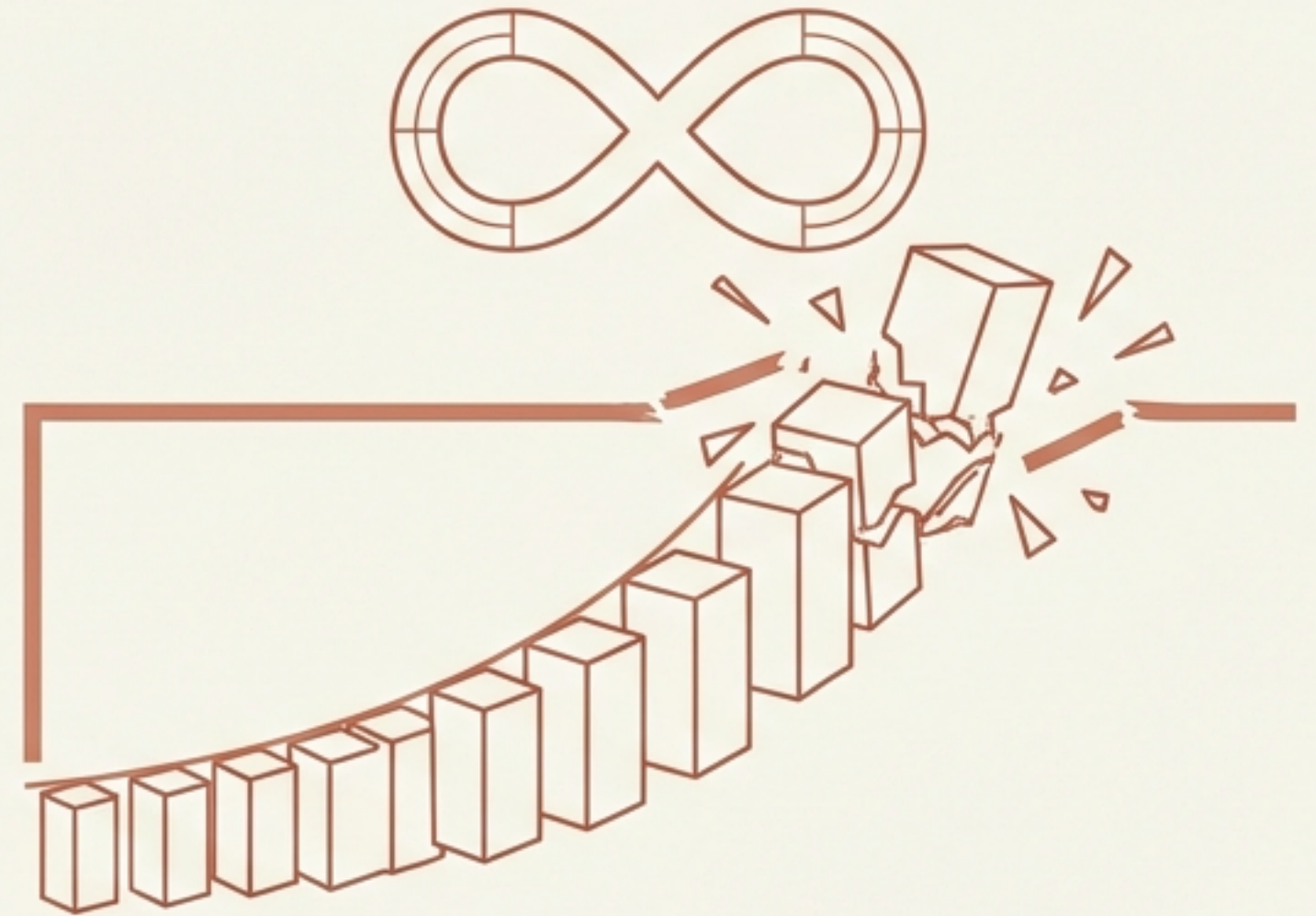


CONVERGENT SERIES



The sequence of partial sums approaches a finite, fixed boundary. The terms shrink fast enough that their infinite sum is a real, measurable number.

DIVERGENT SERIES



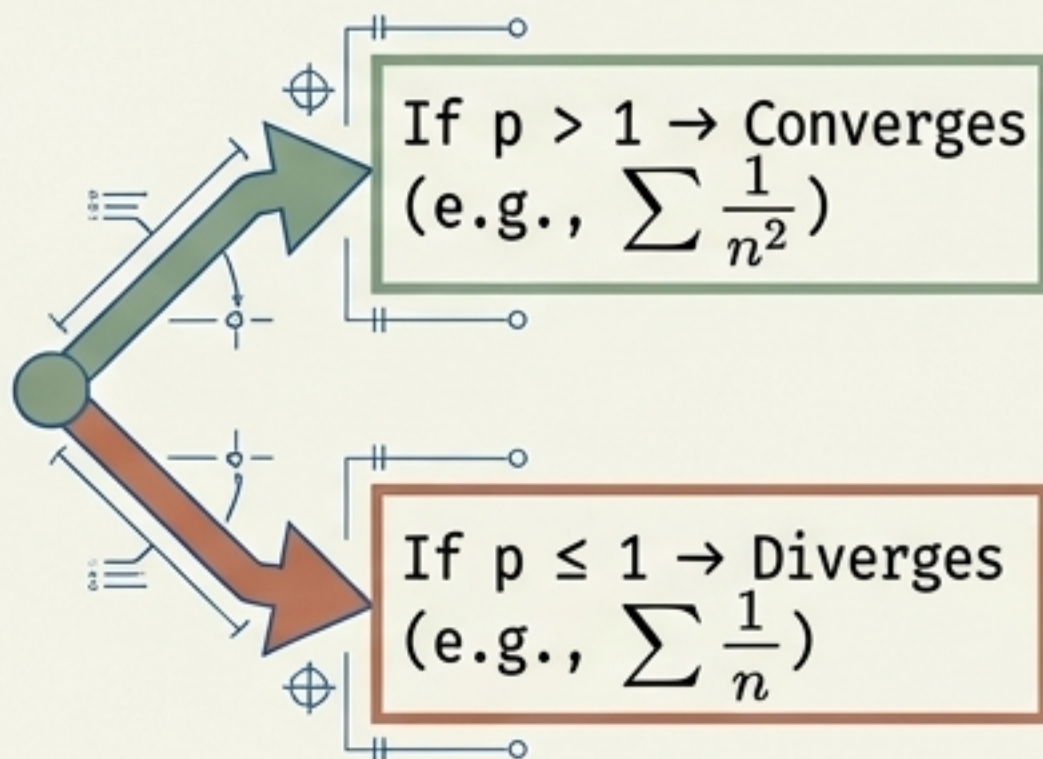
The sum escapes all finite bounds. The terms do not shrink fast enough, driving the total value to infinity or endless oscillation.

The Foundational Yardsticks

Every advanced test ultimately compares an unknown series back to one of these two known truths.

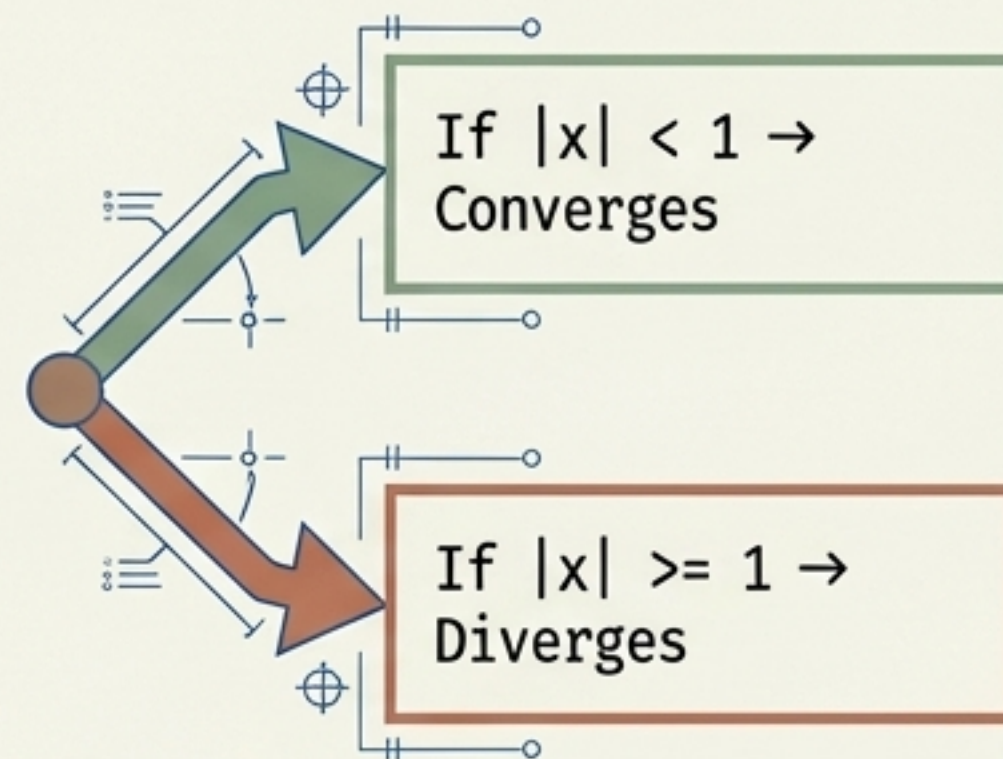
1. The P-Series Yardstick

$$\sum \frac{1}{n^p}$$



2. The Geometric (GP) Yardstick

$$\sum x^n$$



THE PRIME DIRECTIVE: Always identify 'p' or 'x' first. These two foundational truths are the absolute baselines for the Comparison Test.

The Master Convergence Decision Tree

Write the n -th term (u_n)

Yes

LEIBNITZ TEST

Are terms alternating
(+ - + -)?

No

Is the entire term raised
to the power of n ?

Yes

CAUCHY'S ROOT TEST

No

Does u_n contain factorials ($n!$) or
constants to the power of n (x^n)?

Yes

D'ALEMBERT'S RATIO TEST

No
(Mostly polynomials)

COMPARISON TEST

If Limit = 1

RAABE'S TEST

The Comparison Test: Constructing v_n via Degree Extraction

1. Isolate Highest Powers

$$u_n = \frac{(n+1)(n+4)}{n(n+2)(n+5)}$$

Numerator:
 $n \cdot n = n^2$

Denominator:
 $n \cdot n \cdot n = n^3$

2. Calculate the Difference

$$\begin{aligned} \text{Degree} &= \text{Denominator } (n^3) \\ &- \text{Numerator } (n^2) \\ &= n^1 \end{aligned}$$

3. Construct Auxiliary Series

$$v_n = \frac{1}{n^1}$$

4. The Verdict

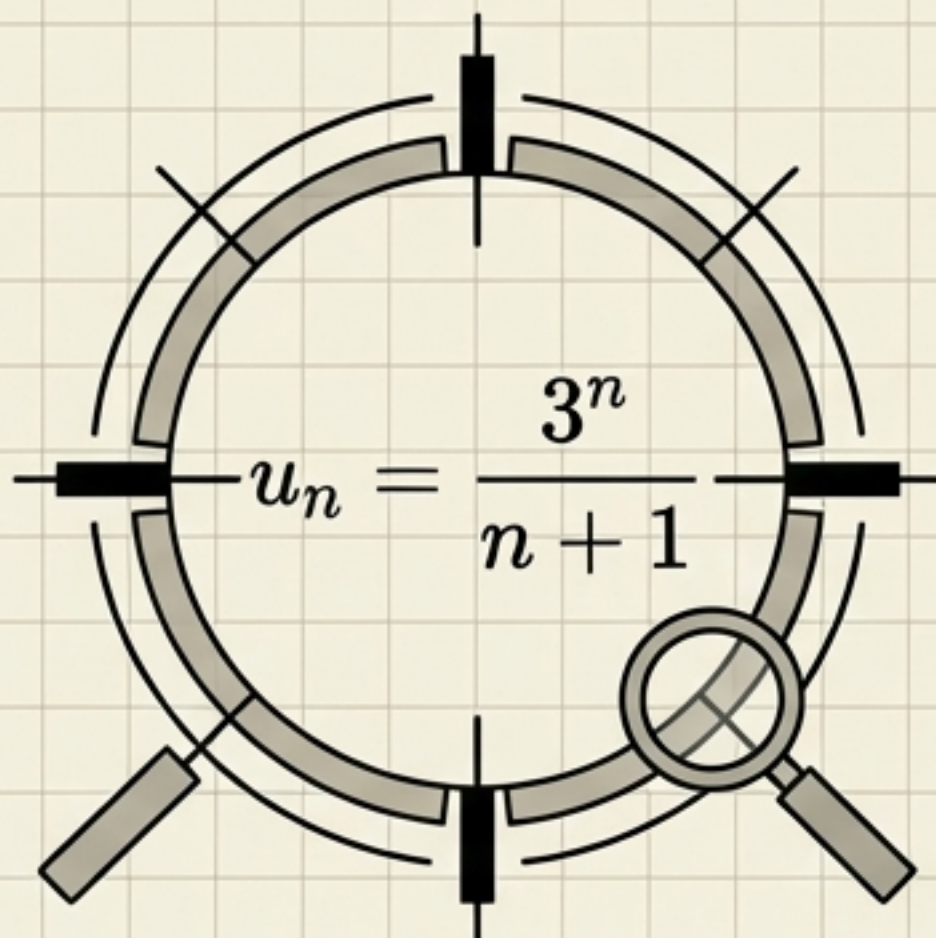
Since $v_n = \frac{1}{n^1}$ is a P-Series with $p=1$, it **DIVERGES**.



Therefore, u_n perfectly mimics v_n and also **DIVERGES**.

D'Alembert's Ratio Test: The Factorial & Exponential Specialist

The Cue



Visual Cue: Look for factorials ($n!$) or constants raised to the power of n . Comparison tests will fail here.

The Mechanism

$$\text{Evaluate } L = \lim_{n \rightarrow \infty} \left| \frac{u_{n+1}}{u_n} \right|$$

Setting up this ratio structurally forces massive factorial and exponential terms to neatly cancel out.

The Verdict Matrix

$$L < 1$$



Converges

$$L > 1$$



Diverges

$$L = 1$$



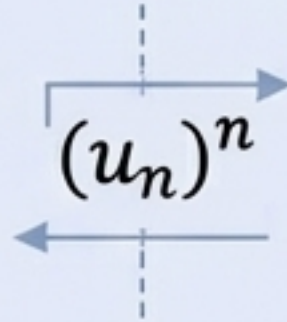
Test Fails
(Initiate Raabe's Test protocols)

Specialized Tactical Protocols

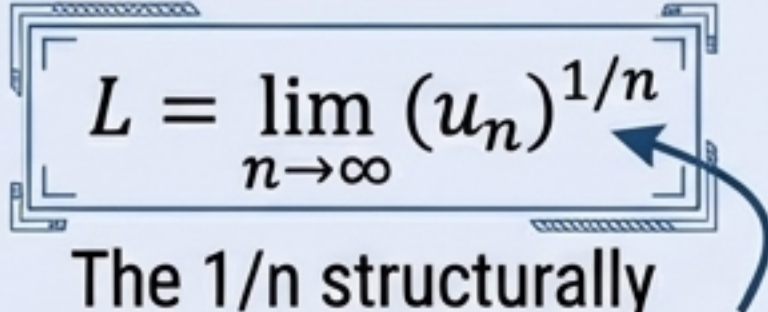
Protocol Alpha: Cauchy's Root Test

The Cue

Deployed when the entire n-th term is raised to the power of n.


$$(u_n)^n$$

The Tool


$$L = \lim_{n \rightarrow \infty} (u_n)^{1/n}$$

The $1/n$ structurally neutralizes the exponent.

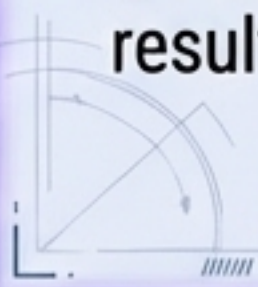
The Rules

- ✓ $L < 1$: Converges
- ✗ $L > 1$: Diverges

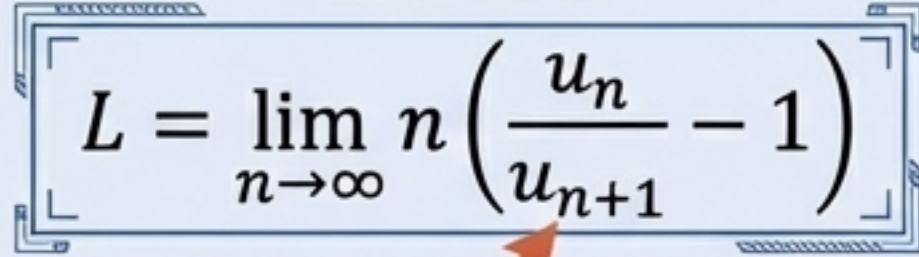
Protocol Beta: Raabe's Test

The Cue

Deployed strictly when **D'Alembert's** Ratio Test results in exactly $L = 1$.


$$L = \frac{3^n}{n + 1}$$

The Tool


$$L = \lim_{n \rightarrow \infty} n \left(\frac{u_n}{u_{n+1}} - 1 \right)$$



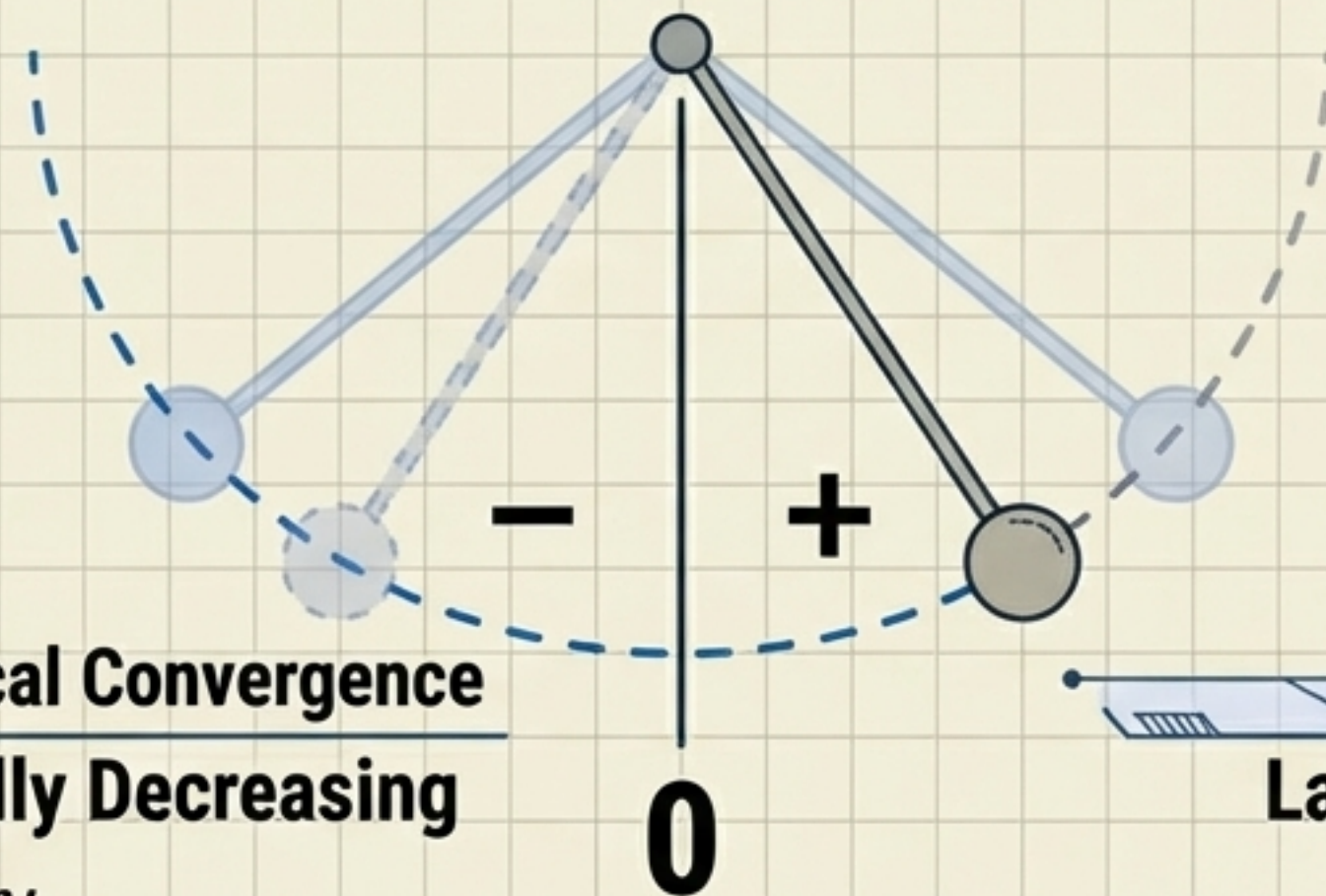
Notice the inversion: u_n is now in the **numerator**.

The Rules

Notice the inverted logic:

- ✓ $L > 1$: Converges
- ✗ $L < 1$: Diverges

Alternating Series & The Leibnitz Pendulum



The Two Laws of Physical Convergence

Law 1: Monotonically Decreasing

$$u_{n+1} \leq u_n$$

Each swing must be strictly shorter than the last. The pendulum must constantly lose momentum.

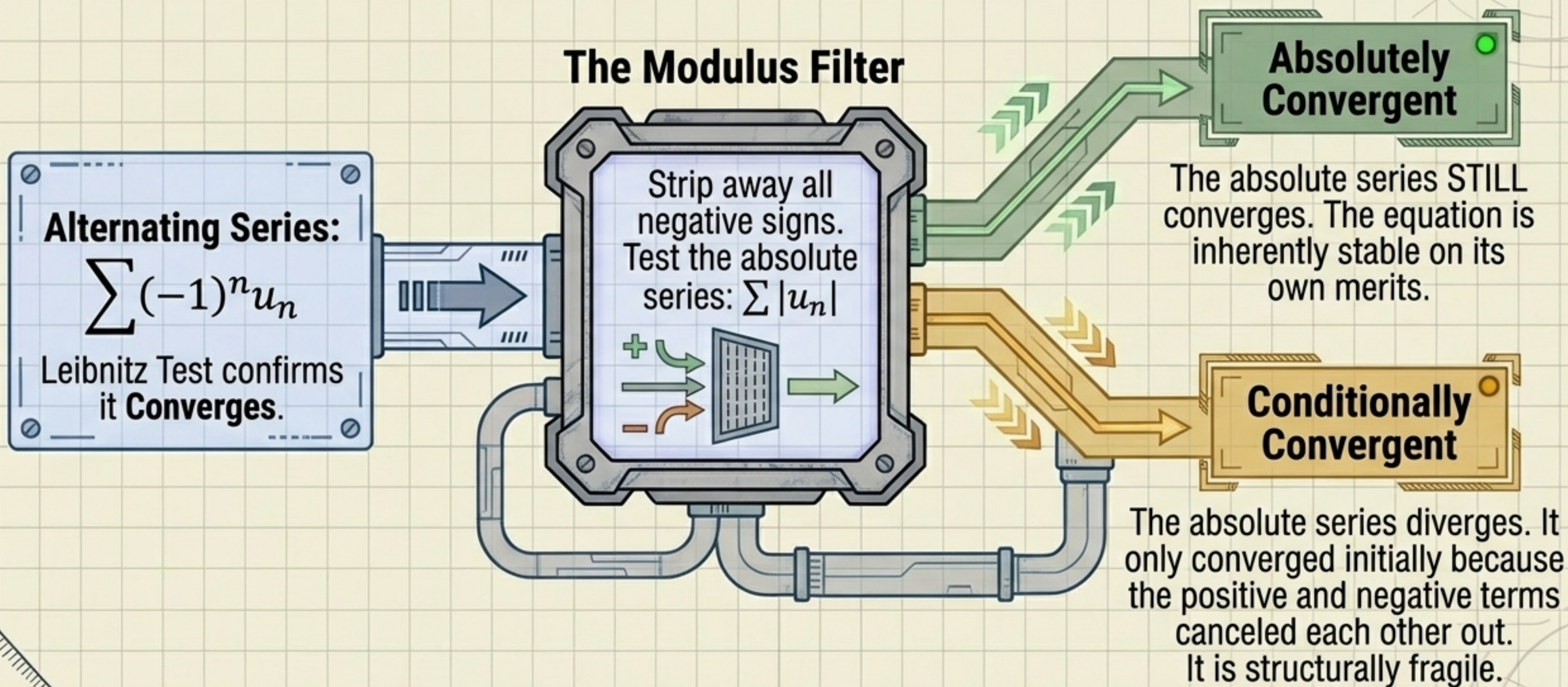
Law 2: Limit to Zero

$$\lim_{n \rightarrow \infty} u_n = 0$$

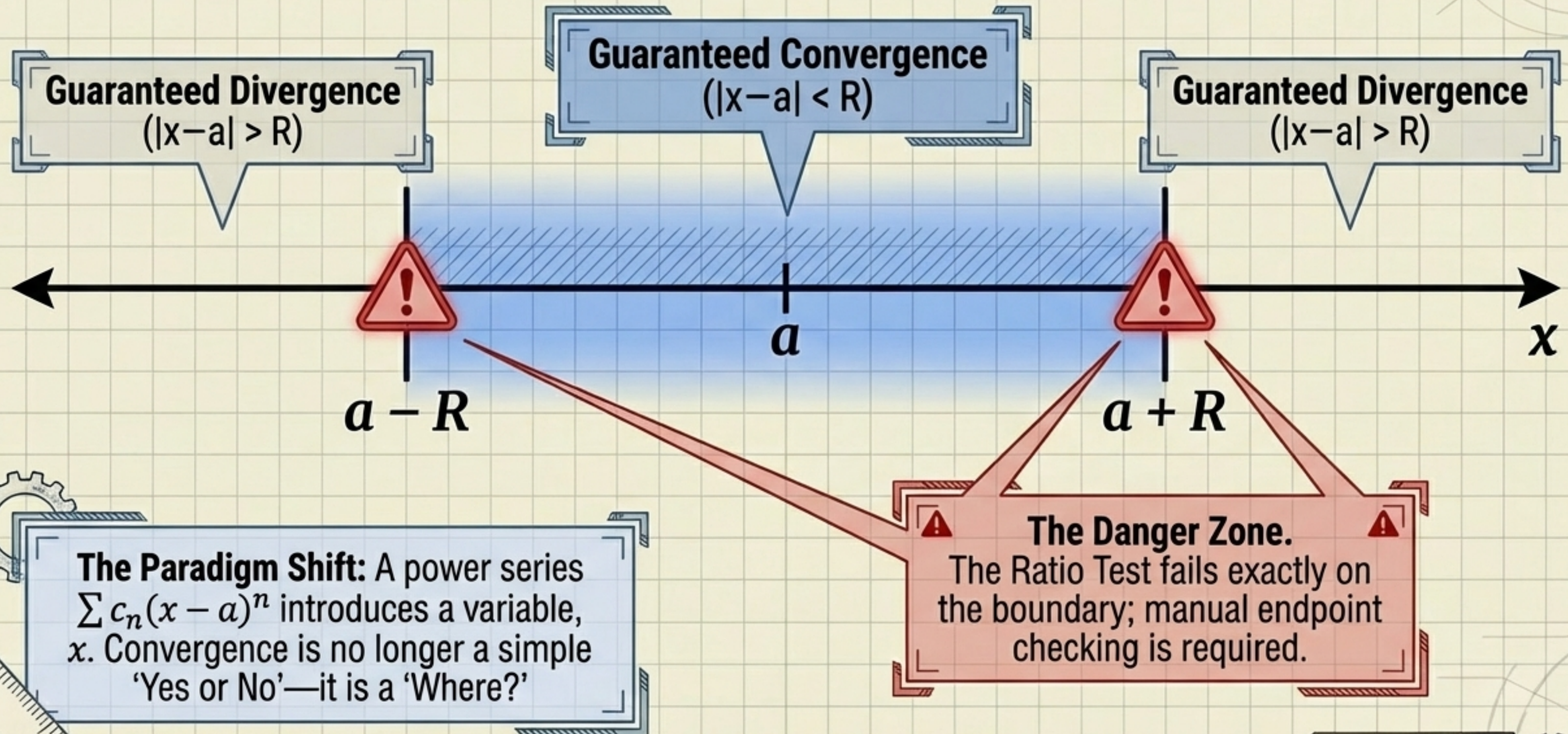
The swings must eventually decay entirely to a resting state at exactly zero.

Verdict: If both laws hold true, the alternating series is securely locked into convergence.

The Modulus Check: Absolute vs. Conditional



Power Series: Discovering the Radius of Convergence



Anatomy of a Taylor Series

1 The Shape Capturer.

The derivatives at point 'a' extract the precise curvature, slope, and concavity of the original function.

3 The Projection.

The polynomial building blocks that push the approximation outward from the center anchor point 'a'.

$$f(x) = \sum \left[\frac{f^{(n)}(a)}{n!} \right] * (x - a)^n$$

2

The Stabilizer.

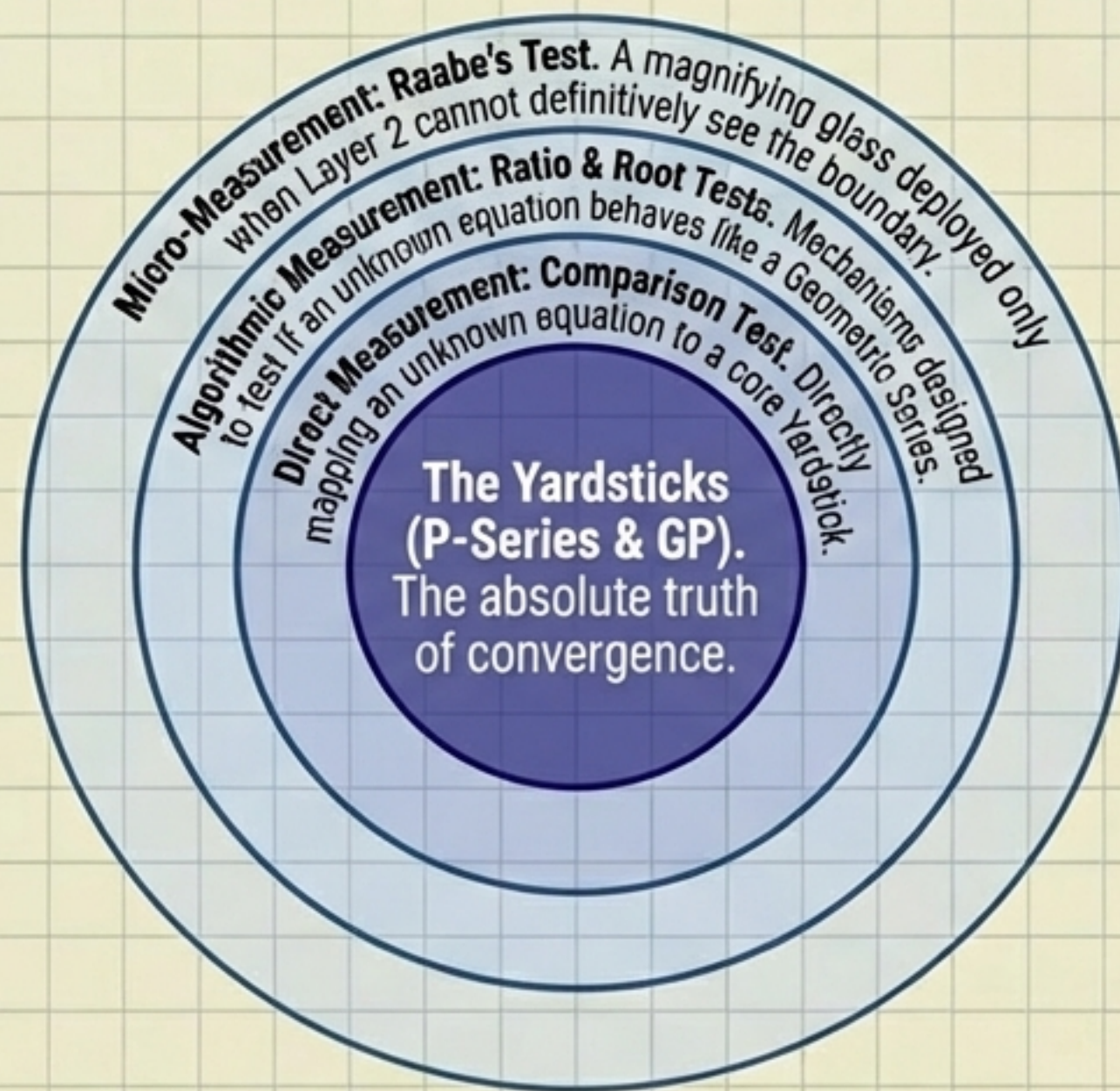
Factorials grow massively in the denominator, suppressing higher-degree terms so the series approximation doesn't wildly blow up to infinity.

$$\sin x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$$

The Standard Expansions Matrix

Exponential	Trigonometric	Logarithmic
$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$ Domain of Convergence: Valid for all x ($-\infty < x < \infty$).	$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$ $\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots$ Domain of Convergence: Valid for all x. (Notice the strictly alternating signs).	$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$ Domain of Convergence: Strictly bounded. Valid only for $-1 < x \leq 1$. ⚠ Notice the distinct lack of factorials in the denominator, reducing its stability range. ⚠

The Convergence Hierarchy



Synthesis Takeaway: Mathematical convergence is not a list of disconnected rules. It is a single, unified system of comparing the unknown against the known.